

This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options.

1 point

- ☐ The set $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly dependent.
- ☐ The set $\{(1, 2, 3), (4, 5, 6), (5, 7, 9)\}$ is linearly independent.
- ☐ The set $\{(1, 2, 3), (0, 5, 6), (0, 0, 9)\}$ is linearly independent.
- ☐ The set $\{(1, 2, 3), (0, 0, 6), (0, 0, 9)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly dependent.
 The set $\{(1, 2, 3), (0, 5, 6), (0, 0, 9)\}$ is linearly independent.

2) If $ad - bc = 0$ then which of the following options are true?

1 point

- ☐ The set $\{(a, b), (c, d)\}$ is linearly dependent in $\mathbb{R}^2$.
- ☐ The set $\{(a, c), (b, d)\}$ is linearly dependent in $\mathbb{R}^2$.
- ☐ The set $\{(1, 0, 0), (0, a, b), (0, c, d)\}$ is linearly dependent in $\mathbb{R}^3$.
- ☐ The set $\{(1, 0, 0), (0, a, c), (0, b, d)\}$ is linearly dependent in $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(a, b), (c, d)\}$ is linearly dependent in $\mathbb{R}^2$.
 The set $\{(a, c), (b, d)\}$ is linearly dependent in $\mathbb{R}^2$.
 The set $\{(1, 0, 0), (0, a, b), (0, c, d)\}$ is linearly dependent in $\mathbb{R}^3$.
 The set $\{(1, 0, 0), (0, a, c), (0, b, d)\}$ is linearly dependent in $\mathbb{R}^3$.

3) For $c \in \mathbb{Z}$, consider the set of three vectors $S = \{(1, c, 0), (c, 0, c), (2c, 3c, 4c)\}$ in $\mathbb{R}^3$ with usual addition and scalar multiplication. Find the value of c such that the given set is linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

4) Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly independent and **1 point** consider the following system of linear equations

$$\begin{aligned}a_1x + b_1y + c_1z &= 1 \\a_2x + b_2y + c_2z &= 2 \\a_3x + b_3y + c_3z &= 3.\end{aligned}$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then

which of the following option(s) is(are) true?

- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐ The system of linear equations have infinitely many solutions.
- ☐ The coefficient matrix A is invertible.
- ☐ The coefficient matrix A is not invertible.
- ☐ Any one of the above options cannot be concluded using the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has a unique solution.
The coefficient matrix A is invertible.

5)

1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly dependent and consider the following system of linear equations

$$a_1x + b_1y + c_1z = 1$$

$$a_2x + b_2y + c_2z = 2$$

$$a_3x + b_2y + c_3z = 3.$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then

which of the following option(s) is(are) true?

- ☐ Either the system of linear equations has a unique solution or no solution.
- ☐ Either the system of linear equations has no solution or infinitely many solutions.
- ☐ Either the system of linear equations have infinitely many solutions or a unique solution.
- ☐ The system of linear equations has a unique solution.
- ☐ The coefficient matrix A is invertible.
- ☐ The coefficient matrix A is not invertible.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Either the system of linear equations has no solution or infinitely many solutions.

The coefficient matrix A is not invertible.

6) If S is a linearly independent set in vector space $\mathbb{R}^3$ with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of S .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point**Level 2**

7) Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of $\mathbb{R}^2$. Choose the set of correct options. **1 point**

- ☐ $\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- ☐ $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- ☐ $\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- ☐ $\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
 $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of $\mathbb{R}^3$.

8) Each column of an $n \times n$ matrix A can be thought of as vectors in $\mathbb{R}^n$. If a set S consists of all the column vectors of A , then which of the following is(are) true? **1 point**

- ☐ S is always linearly independent.
- ☐ S is linearly independent if $\det(A) = 0$.
- ☐ S is linearly independent if $\det(A) = 0$.
- ☐ If $\det(A) = 1$, then S is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent if $\det(A) = 0$.
 If $\det(A) = 1$, then S is linearly independent.

9) Suppose it is known that the three vectors $v_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, $v_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$, $v_3 = \begin{bmatrix} g \\ h \\ i \end{bmatrix}$ are linearly dependent in $\mathbb{R}^3$. Which of the following is correct? **1 point**

- ☐ Some linear combination of a, b, c must be 0.
- ☐ Some linear combination of a, d, g must be 0.
- ☐ Some linear combination of d, e, f must be 0.
- ☐ Some linear combination of c, f, i must be 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Some linear combination of a, b, c must be 0.
 Some linear combination of a, d, g must be 0.
 Some linear combination of d, e, f must be 0.
 Some linear combination of c, f, i must be 0.

10) If vectors $(2x, 4, -6)$, $(-6, -3x, 2x)$, $(-1, x - 3, x + 2)$ from the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$, then find the value of x .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Your score is: 0/10